

Theorem (Markov's inequality). *Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let $X : \Omega \rightarrow [0, \infty]$ be a non-negative random variable. For any real number $a > 0$,*

$$\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}[X]}{a}.$$

Moreover,

$$\mathbb{P}(X \geq a) = \frac{\mathbb{E}[X]}{a} \iff \mathbb{P}(X \in \{0, a\}) = 1.$$

Proof. 1. **Notation.** For the fixed $a > 0$ define the indicator of the event $\{X \geq a\}$ by

$$\mathbf{1}_{\{X \geq a\}}(\omega) = \begin{cases} 1, & X(\omega) \geq a, \\ 0, & X(\omega) < a. \end{cases}$$

2. **Pointwise elementary inequality.** For every $\omega \in \Omega$,

$$a \mathbf{1}_{\{X \geq a\}}(\omega) \leq X(\omega). \quad (2.1)$$

Indeed, if $X(\omega) \geq a$ the left-hand side equals $a \leq X(\omega)$; if $X(\omega) < a$ the left-hand side is $0 \leq X(\omega)$ because $X \geq 0$.

3. **Take expectations.** Both sides of (2.1) are non-negative, so by monotonicity of expectation (if $Y \leq Z$ a.s. then $\mathbb{E}[Y] \leq \mathbb{E}[Z]$) we obtain

$$a \mathbb{E}[\mathbf{1}_{\{X \geq a\}}] \leq \mathbb{E}[X]. \quad (3.1)$$

Linearity of expectation allows pulling the constant a out.

4. **Identify the expectation of an indicator.** For any event A , $\mathbb{E}[\mathbf{1}_A] = \mathbb{P}(A)$. Hence

$$a \mathbb{P}(X \geq a) \leq \mathbb{E}[X]. \quad (4.1)$$

5. **Divide by $a > 0$.** Since a is positive,

$$\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}[X]}{a},$$

which is Markov's inequality.

6. **Equality condition.** Equality in the final inequality can occur only if equality holds in every preceding step.

- *Equality in (2.1).* Equality a.s. in (2.1) means

$$a \mathbf{1}_{\{X \geq a\}}(\omega) = X(\omega) \quad \text{for } \mathbb{P}\text{-a.e. } \omega,$$

i.e.

$$X(\omega) = \begin{cases} a, & X(\omega) \geq a, \\ 0, & X(\omega) < a. \end{cases} \quad (6.1)$$

Thus X takes only the two values 0 and a (almost surely).

- *Equality in the monotonicity step (??).* The inequality $\mathbb{E}[Y] \leq \mathbb{E}[Z]$ is an equality iff $Y = Z$ $\mathbb{P}$ -a.s.; hence (??) is an equality precisely when (??) holds, which is the same condition obtained from (??).

Consequently,

$$\mathbb{P}(X \geq a) = \frac{\mathbb{E}[X]}{a} \iff \mathbb{P}(X \in \{0, a\}) = 1.$$

7. **Conclusion.** For any non-negative random variable X and any $a > 0$,

$$\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}[X]}{a},$$

with equality exactly when X assumes only the values 0 and a almost surely. $\square$

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